Ahamed Mohamed Wajibu

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EDUCATION

UNIVERSITY OF WATERLOO — Waterloo, ON
Bachelor of Mathematics — May 2028

SKILLS / TECHNOLOGIES

- **Languages:** Python, C++, Rust, Go, C#, TypeScript, JavaScript, C
- **ML & Inference:** CUDA, cuBLAS, PyTorch, TensorFlow, LangChain, LLM Inference
- **Cloud & Developer Tools:** Azure, AWS, Docker, Linux, Git, Microsoft Fabric
- **Frameworks / etc.:** .NET / ASP.NET Core, Flask, FastAPI, Node.js, React, SQL Server, PostgreSQL, OpenCV, Nginx, Unit Testing

EXPERIENCE

RBC Borealis — Toronto, ON
AI Researcher Fellow — *October 2026 – December 2026*

- Developing a computer vision pipeline using **Python**, **YOLO**, and **ByteTrack** / **DeepSORT** to detect and track players and helmets and attribute head impacts from multi-camera football footage.
- Building **PyTorch temporal models** over player-tracking features including relative velocity, acceleration, collision angle, and player orientation to classify impacts and aggregate per-athlete exposure profiles.

Mitsubishi — Mississauga, ON
AI Engineer Intern — *January 2026 – May 2026*

- Built an internal **RAG platform** for aircraft technical manuals using **LangChain, Azure AI Foundry, Ollama, PostgreSQL** and **Microsoft Fabric** supporting **40** engineers at deployment via an internal AI chatbot.
- Built document-processing pipelines for **2,000+** aircraft manuals, extracting structured fields, per-page summaries, and source metadata into **PostgreSQL**.
- Deployed on-prem **Ollama** inference **APIs** behind **Nginx reverse proxy**, serving Gemma and quantized open-source models for secure internal document analysis and retrieval.
- Integrated **tool calling** into the internal **AI assistant**, enabling document search, page retrieval, **PostgreSQL** keyword search, web search, and inline source citations.

Mitsubishi — Mississauga, ON
Software Developer Intern — *September 2025 – December 2025*

- Built **ASP.NET Core / C# pages** replacing manual **SQL Server** scripts for QA call-up updates, aircraft line status updates, job updates, and Syspro service workflows.
- Developed **SQL scripts** and **ETL** data workflows for production data fixes, validation checks, employee/job records, and manufacturing system updates.
- Automated repetitive **Syspro** and production support workflows for **100s** of production workers and dozens of **Syspro** users, reducing manual staff time spent on database update requests.
- Refactored **Entity Framework** services**, LINQ** queries**,** and **SQL Server** access patterns, cutting request latency by **~30%** and reducing redundant database calls across production modules.

PROJECTS

Forge | *Rust, C++, CUDA, Go* | *Check it out ↗*

- Built an **LLM inference engine** from scratch for Qwen2.5-0.5B, implementing **transformer decoding**, **KV caching**, **RoPE**, RMSNorm, and attention kernels in C++ with a Rust runtime and Go HTTP API.
- Optimized CPU inference from **0.44 to 23 tokens/sec (~52× speedup)** through multithreading, **INT8/INT4 weight quantization**, fused kernels, and memory optimizations.
- Implemented a **CUDA** inference backend with custom GPU kernels, **cuBLAS** acceleration, device-resident KV caching, and continuous batching to serve concurrent requests.

Block-GTQ WebGPU | *TypeScript, WGSL, WebGPU, PyTorch* | *Check it out ↗*

- Ported **Block-GTQ**'s **RoPE**-aware **KV-cache** quantization to **WebGPU/WGSL**, implementing mixed-width encoding, bit-packed storage, and compressed attention for browser LLM inference.
- Integrated compressed **K caching** into SmolLM2-135M, reducing persistent K memory **2.24×** at **4K** context.
- Reworked compressed attention to avoid full-K reconstruction, reducing scratch buffers from **2.20 MB** to **101 KB** and cutting attention latency by **22×** versus the initial compressed implementation on a **4K×128** synthetic workload.